

STQFORMER: SPATIO-TEMPORAL QUATERNION TRANSFORMER FOR VIDEO FRAME DENOISING

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ABSTRACT

Preserving subtle spectral variations and temporal consistency is critical for reliable endoscopic video denoising. However, most existing methods process RGB channels independently and emphasize spatial noise suppression, often causing color distortion and temporal incoherence. In this paper, we propose **STQFormer**, a spatio-temporal denoising framework that models color channels as a unified vector entity and captures long-range temporal dependencies via quaternion-based representations. STQFormer integrates Spatio-Temporal Quaternion Transformer blocks with a frame-wise Swin encoder and is trained in an unsupervised manner without paired clean-noisy data. Experimental results on endoscopic video datasets demonstrate that STQFormer achieves state-of-the-art denoising performance while preserving diagnostic color fidelity and temporal coherence.

Index Terms— Video Denoising, Spatio-Temporal Modeling, Transformer, Quaternion Convolution, Medical Imaging

1 Introduction

Video denoising is a fundamental task in low-level vision, aiming to suppress noise while preserving spatial structures and temporal consistency. In endoscopic video analysis, this problem is particularly challenging, as subtle color variations and fine-grained temporal cues are critical for clinical diagnosis but highly sensitive to noise and illumination variations. Moreover, the scarcity of paired clean-noisy medical videos makes unsupervised learning a practical necessity in real-world clinical settings.

Recent deep learning-based methods have advanced video restoration but face specific limitations in endoscopy. Transformer-based architectures, such as Swin-based models [1] and Restormer [2], improve representation capacity but primarily focus on spatial modeling. They treat RGB channels independently, overlooking intrinsic inter-channel spectral correlations crucial for diagnostic color fidelity. Unsupervised methods like TAP [3] leverage pre-trained priors but suffer from domain gaps when applied to endoscopic tissue textures. Conversely, while medical-specific frameworks

like VSRM [4] demonstrate effective spatio-temporal modeling, they rely on paired supervision, limiting their utility in clinical settings where clean data is scarce.

Motivated by these limitations, we propose STQFormer, a spatio-temporal denoising framework tailored for unsupervised endoscopic video denoising. Our main contributions are summarized as follows:

- We introduce STQFormer as the first unsupervised framework explicitly designed to preserve both diagnostic color fidelity and temporal coherence in endoscopic videos.
- We introduce the Spatio-Temporal Quaternion Transformer (STQT), which models RGB channels as a unified vector entity to preserve inter-channel spectral correlations while efficiently capturing long-range temporal dependencies.
- We design a curriculum-based two-stage training strategy that stabilizes unsupervised optimization by progressively transitioning from spatial reconstruction to full spatio-temporal modeling.
- We validate STQFormer on the Kvasir-Capsule dataset [5], achieving state-of-the-art results for unsupervised denoising with a real-time inference speed of 18.6 FPS, demonstrating its strong clinical utility.

2 Related Work

2.1 Video Denoising Paradigm

Video denoising has evolved from early motion-based filtering to deep learning frameworks. **Supervised Approaches:** Methods using 3D convolutions [6] or Transformers [1, 2] exploit temporal redundancy and have set benchmarks on generic datasets. Recently, VSRM [4] introduced state-space models to medical video restoration. However, these methods fundamentally rely on paired clean-noisy data. In endoscopy, acquiring ground truth is clinically impractical, and models trained on synthetic data often generalize poorly to real-world tissue artifacts.

Unsupervised Approaches: To address data scarcity, unsupervised frameworks like TAP [3] leverage pre-trained image denoisers as spatial priors. While effective for natural scenes, they treat RGB channels independently and rely on generic priors that misinterpret subtle biological textures as noise (domain gap). Unlike these methods, our STQFormer is explicitly designed for unsupervised learning on medical data,

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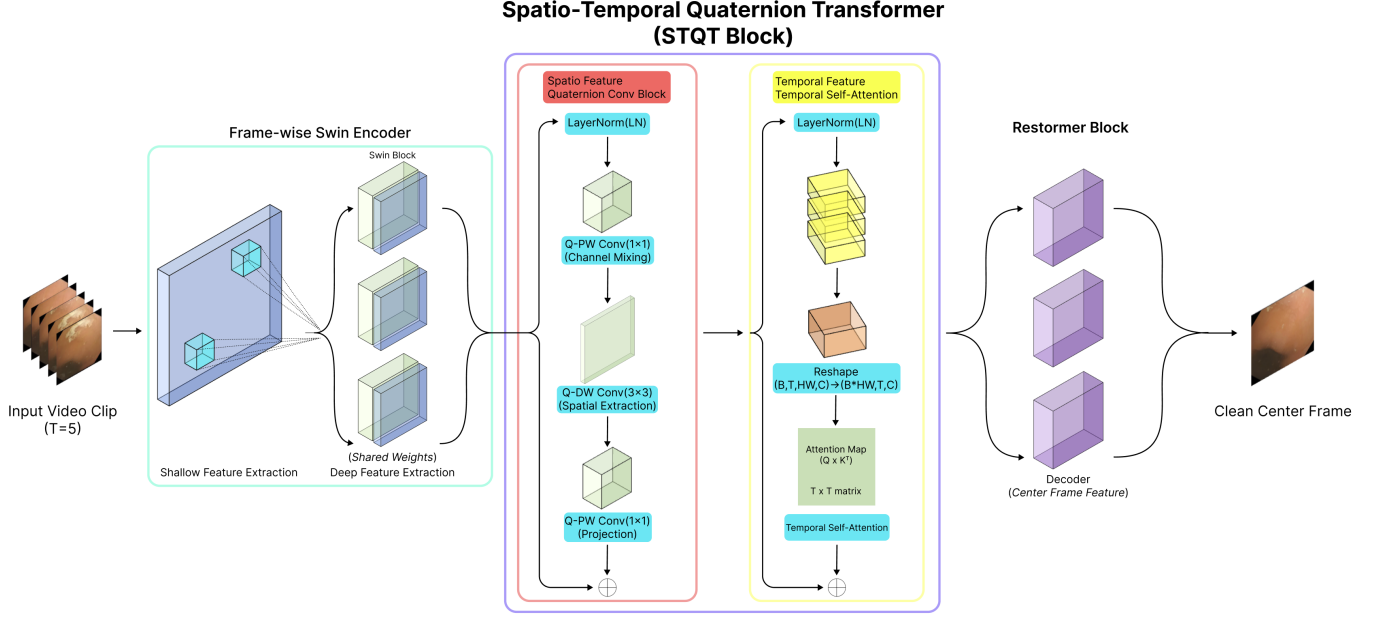


Fig. 1. STQFormer Architecture. A weight-shared **Frame-wise Swin Encoder** extracts features from a noisy clip ($T = 5$). **Spatio-Temporal Quaternion Transformer (STQT)** blocks then decouple spatial (Quaternion Conv) and temporal (Self-Attention) modeling. Finally, a **Restormer-based Decoder** predicts the clean center frame.

bypassing the need for paired supervision or generic priors.

2.2 Quaternion-Based Representation

Quaternion algebra offers a robust framework for processing multi-dimensional signals by treating color channels as unified vector entities rather than independent scalars. Quaternion Neural Networks (QNNs) have demonstrated superior color preservation with fewer parameters compared to real-valued counterparts [7]. Recently, quaternion-based mechanisms have been explored for single image tasks [8]. However, their potential in *spatio-temporal* video modeling remains unexplored.

3 Proposed Method

3.1 Overall Architecture

Given a sequence of noisy frames

$$X = \{I_{t-T/2}, \dots, I_t, \dots, I_{t+T/2}\}, \quad (1)$$

where X denotes the input video clip consisting of T consecutive frames ($T = 5$ in our implementation), and $I_t \in \mathbb{R}^{3 \times H \times W}$ represents the noisy center frame at time step t , with H and W denoting its height and width. This center frame I_t serves as the primary target for restoration. STQFormer predicts a clean version of the center frame I_t . As illustrated in Fig. 1, the architecture consists of three main components: a Frame-wise Encoder, Spatio-Temporal Feature Modeling, and a Center-Frame Decoder.

3.2 Frame-wise Encoder

We adopt a pre-trained Swin Transformer [1] as the backbone encoder. Each frame is independently encoded into a feature map $F_t = E(I_t)$. To align feature dimensions, a 1×1 convolution projects encoder outputs to a unified embedding dimension C .

3.3 Spatio-Temporal Quaternion Transformer (STQT)

The core of our method is the STQT block. To bypass the error propagation risks of explicit alignment modules (e.g., optical flow) [3] and the computational prohibitive nature of full spatio-temporal attention, STQT adopts a decoupled modeling strategy. This design allows the model to implicitly aggregate coherent biological features across frames via attention, ensuring robust restoration without relying on noisy physical alignment, while significantly improving computational efficiency.

3.3.1 Spatial Modeling (Quaternion Block)

For spatial feature extraction, we represent color-aware features in the quaternion domain. Specifically, the RGB input is mapped to a pure quaternion $q = 0 + Ri + Gj + Bk$, where the real part is set to zero and the three imaginary components correspond to the R, G, and B channels. For intermediate feature maps, channels are grouped into quaternion units of four components, enabling joint modeling of cross-channel spectral correlations.

In implementation, we adopt an efficient quaternion-

inspired depthwise-separable block:

$$F_{\text{spatial}} = \text{PW}_2(\text{DW}(\text{PW}_1(F_{\text{in}}))). \quad (2)$$

Here, PW_1 performs cross-component mixing among quaternion feature groups, DW extracts spatial structures, and PW_2 projects the features back to the target embedding dimension. This design preserves inter-channel color relationships while maintaining computational efficiency.

3.3.2 Temporal Modeling

To capture temporal dependencies, we perform multi-head self-attention along the temporal dimension. Given the spatially enhanced features, we reshape them to $(B \cdot H \cdot W, T, C)$ and compute standard scaled dot-product attention using projections Q, K, V . This allows the model to selectively aggregate information from neighboring frames to refine the center frame.

3.4 Decoder and Center-Frame Prediction

After spatio-temporal modeling, only the center frame feature is decoded using a Restormer-style [2] decoder. This design avoids supervision misalignment and focuses the model’s capacity on the target frame I_t .

3.5 Unsupervised Objective and Strategy

To enable robust training without paired clean–noisy data, we propose a **Stochastic Neighbor Loss** inspired by Noise2Noise [9], integrated with a **Curriculum Strategy** [10].

Instead of a clean ground truth, the supervision target I_{tgt} for the center frame is randomly sampled from its temporal neighborhood $(t \pm k, k \in \{1, 2\})$. Leveraging the high temporal redundancy of endoscopic streams, we assume local structural consistency within this small window despite tissue deformation. This stochastic selection approximates the clean signal expectation under the zero-mean noise assumption, helping separate noise from the underlying biological structure. The total objective is

$$\mathcal{L}_{\text{total}} = \lambda_{\text{pix}} \mathcal{L}_{\text{char}}(\hat{I}_t, I_{tgt}) + \lambda_{\text{ssim}} \mathcal{L}_{\text{ssim}} + \lambda_{\text{perc}} \mathcal{L}_{\text{perc}} + \lambda_{\text{tc}} \mathcal{L}_{\text{tc}}. \quad (3)$$

Here, we employ Charbonnier loss [11], SSIM loss [12], and Perceptual loss [13] for spatial fidelity. Additionally, we incorporate a **Temporal Consistency loss** ($\mathcal{L}_{\text{tc}}$) [14] to penalize temporal flickering between consecutive frames. To stabilize convergence, we optimize the objective in two stages: Stage 1 disables temporal attention and sets $\lambda_{\text{tc}} = 0$ to focus on spatial reconstruction; Stage 2 enables temporal attention and activates $\mathcal{L}_{\text{tc}}$ for full spatio-temporal fine-tuning.

Remark on Noise Assumption. Standard unsupervised denoising assumes zero-mean noise [9], whereas endoscopic videos may contain complex signal-dependent artifacts. Temporal aggregation helps reduce transient noise bias, but highly signal-dependent noise and persistent hardware artifacts remain challenging cases.

4 Experiments

4.1 Dataset and Implementation Details

We evaluate our method on the **Kvasir-Capsule** dataset [5], the largest open-access benchmark for endoscopic video analysis. It contains over 44,000 frames with validated anatomical and pathological variations, serving as a standard for assessing robust performance in complex gastrointestinal environments.

For quantitative evaluation only, we synthesize a paired benchmark by corrupting clean frames with a signal-dependent Poisson–Gaussian noise model:

$$y = x + n(x), \quad \mathbb{E}[n(x)] = 0, \quad \text{Var}[n(x)] = \sigma_s x + \sigma_c^2, \quad (4)$$

where x and y denote the clean and corrupted normalized frames, respectively, and $\sigma_s, \sigma_c \in [0.001, 0.01]$. PSNR and SSIM are then computed between the restored output and the clean frame. Importantly, this synthetic corruption is used to construct a standardized quantitative benchmark and, where required, to retrain supervised comparison models; STQFormer itself is trained fully unsupervised on the original noisy Kvasir-Capsule videos without paired clean–noisy supervision.

Implementation. STQFormer is implemented in PyTorch and trained on 4 NVIDIA RTX 4090 GPUs with batch size 24 and AdamW [15]. Following Sec. 3.5, Stage 1 trains for 8 epochs with temporal attention disabled and the encoder frozen, using a learning rate of 2×10^{-4} ; Stage 2 fine-tunes the full network for 25 epochs with temporal modules enabled and the encoder unfrozen at 1×10^{-4} . The input clip length is $T = 5$, with two neighboring frames on each side.

4.2 Comparison with State-of-the-Arts

We compare our STQFormer with representative methods, including supervised baselines (EDVR [6], Video Swin [1], Restormer [2], BasicVSR [16]) and recent competitors (TAP [3], VSRM [4]). Table 1 reports the restoration quality (PSNR/SSIM) and efficiency metrics (Params, FLOPs, FPS).

Analysis of Competitors:

- **VSRM:** VSRM [4] was originally designed for video super-resolution. For fair denoising comparison, we replace its upsampler with a dimension-preserving head and retrain it under the denoising setting. Although trained with synthetic paired supervision, the adapted VSRM still trails STQFormer by 0.82 dB, suggesting that SR-oriented hallucination is less suitable for noise suppression.
- **TAP:** TAP [3] suffers from a domain gap because its frozen natural-image prior does not adapt well to endoscopic tissue textures, leading to lower PSNR and slower inference.

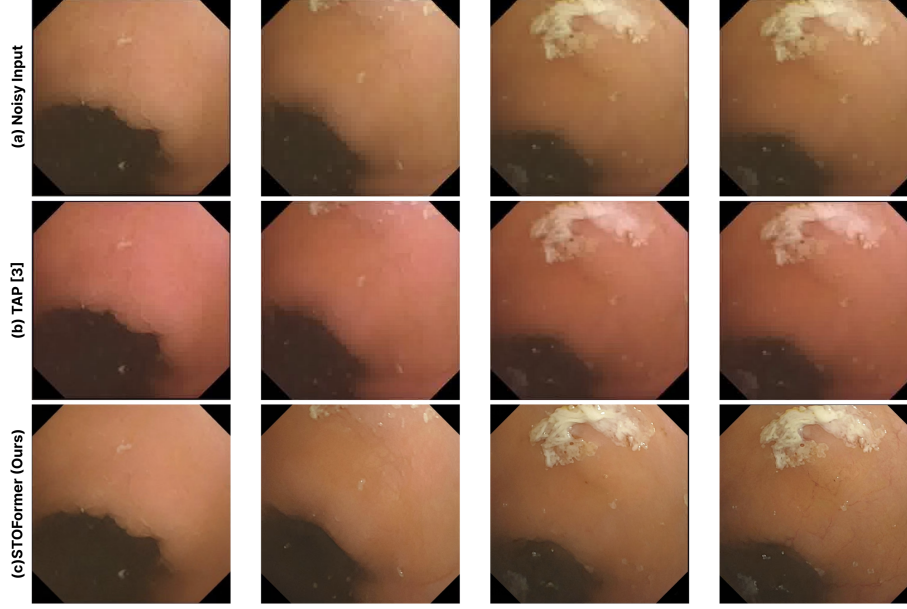


Fig. 2. Qualitative comparison of denoising results on challenging endoscopic video frames. (b) **TAP** [3] suffers from **severe temporal blurring** and **detail loss**. (c) Our **STQFormer** effectively suppresses noise while preserving sharp structural details and color fidelity, demonstrating the robustness of our implicit attention mechanism.

Table 1. Quantitative comparison on the Kvasir-Capsule dataset. We report PSNR, SSIM, parameters in millions (M), computational cost in GFLOPs, and inference speed in frames per second (FPS). **Bold** indicates the best performance, and underline indicates the second best for each metric.

Method	Params (M) ↓	FLOPs (GFLOPs) ↓	FPS (frames/s) ↑	PSNR (dB) ↑	SSIM ↑
<i>Baselines</i>					
EDVR [6]	20.6	190.2	8.5	31.20	0.880
Video Swin [1]	28.3	250.4	6.2	32.50	<u>0.895</u>
Restormer [2]	26.1	144.0	12.1	30.80	0.865
BasicVSR [16]	6.3	45.3	25.0	31.95	0.878
<i>Compare</i>					
TAP [3]	<u>16.5</u>	120.5	5.8	31.45	0.872
VSRM [4]	17.1	127.6	7.5	<u>32.60</u>	0.885
STQFormer (Ours)	23.8	<u>77.5</u>	18.6	33.42	0.918

Overall, STQFormer achieves the best PSNR/SSIM while maintaining 18.6 FPS, showing a favorable trade-off between restoration quality and efficiency.

4.3 Qualitative Analysis

As shown in Figs. 2 and 3, competitor methods suffer from distinct artifacts: **TAP** [3] exhibits severe temporal blurring due to domain gaps, while **VSRM** [4] tends to over-smooth vascular patterns. In contrast, **STQFormer** effectively suppresses noise while preserving sharp diagnostic details. Furthermore, the temporal intensity profiles in Fig. 4 confirm that our method significantly restores temporal coherence compared to the noisy input.

Table 2. Ablation study on key components of STQFormer. **Q-Conv**: Quaternion Convolution; **Temp-Attn**: Temporal Attention; **Strategy**: Two-Stage Curriculum Training. The results demonstrate the progressive improvement of each module.

Model Variant	Q-Conv	Temp-Attn	Strategy	PSNR	SSIM
Baseline (Swin+CNN)	×	×	N/A	30.12	0.852
+ Spatial Q-Conv	✓	×	N/A	31.45	0.871
+ Temporal (Joint)	✓	✓	Joint	32.89	0.905
STQFormer (Full)	✓	✓	2-Stage	33.42	0.918

4.4 Ablation Studies

To systematically analyze the contribution of each component in STQFormer, we conduct ablation experiments on the Kvasir-Capsule dataset [5]. Table 2 summarizes the effects of three key components: Quaternion Convolution (Q-Conv), Temporal Attention (Temp-Attn), and the proposed Two-Stage Training Strategy. We additionally evaluate the temporal target gap used in stochastic neighbor supervision.

The results in Table 2 show that progressively integrating quaternion modeling, temporal dependencies, and curriculum training leads to consistent performance gains.

4.4.1 Effectiveness of Quaternion Convolution

Replacing standard real-valued convolutions with Quaternion Convolution blocks improves PSNR by **1.33 dB**. This is important for endoscopic videos, where subtle color variations reflect tissue properties. By modeling RGB channels as a unified quaternion vector, STQFormer better preserves inter-channel spectral correlations than channel-independent con-

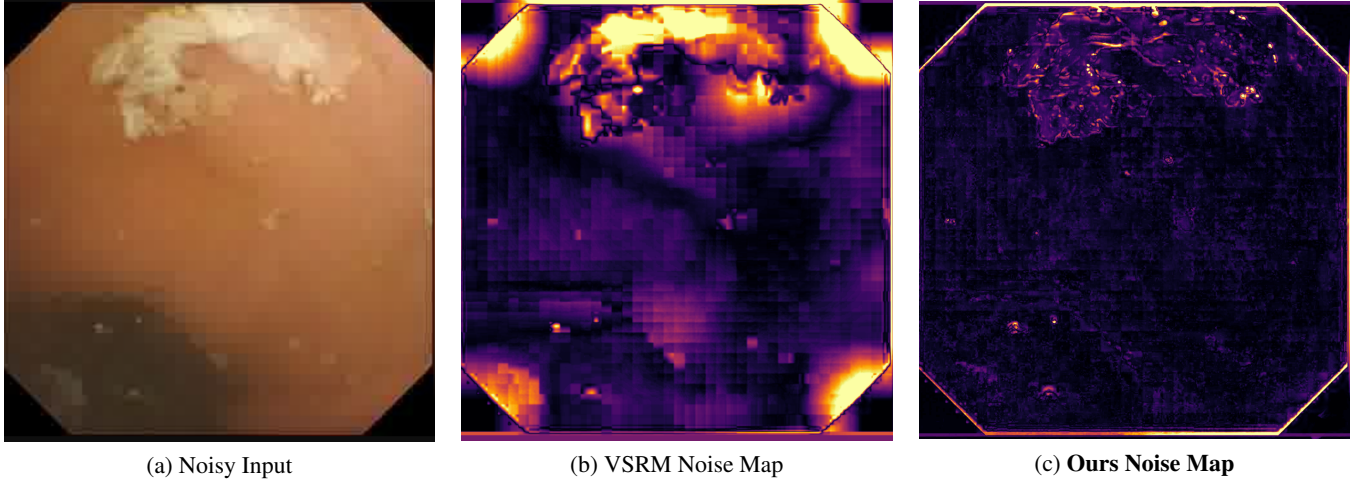


Fig. 3. Visualization of **Method Noise** (absolute difference between noisy input and denoised output). Brighter regions represent the content removed by the model. **(b)** VSRM [4] removes significant structural details, visible as bright vascular patterns in the noise map, suggesting over-smoothing. **(c)** Our STQFormer produces a noise map that contains mostly random noise without anatomical structures, demonstrating superior preservation of diagnostic textures.

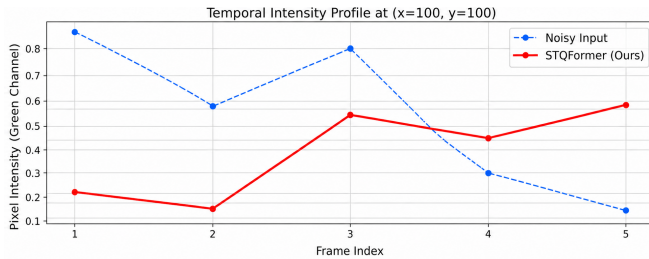


Fig. 4. Analysis of Temporal Consistency. We plot the intensity profile of a single pixel (Green Channel) across consecutive frames. The **Noisy Input** (blue dashed line) exhibits jagged fluctuations indicating severe flickering, whereas **STQFormer** (red solid line) produces a smooth and stable trajectory, demonstrating effective temporal coherence preservation.

volution.

4.4.2 Impact of Temporal Attention

Temporal attention improves PSNR from 31.45 dB to 32.89 dB by aggregating neighboring-frame information at each spatial location. This suppresses temporally inconsistent flickering while reinforcing structures that persist across frames.

4.4.3 Necessity of Two-Stage Training

As shown in the last two rows of Table 2, we compare the standard joint training (Row 3) with our proposed two-stage strategy (Row 4). Although both models utilize the same architecture (Q-Conv + Temporal Attention), directly optimizing the full network from scratch yields suboptimal con-

vergence (32.89 dB). In contrast, our Two-Stage Curriculum Strategy—which aligns spatial features before introducing temporal dynamics—brings an additional gain of 0.53 dB. This validates that progressive training is crucial for stabilizing spatio-temporal feature fusion. **Effect of temporal target gap.** We also test different neighbor gaps for stochastic supervision. The $\pm\{1, 2\}$ setting performs best with 33.42 dB / 0.918, slightly better than ± 1 frames (33.35 dB / 0.915). Larger gaps clearly degrade performance ($\pm\{1, 2, 3, 4\}$: 31.85 dB / 0.880; ± 8 : 29.50 dB / 0.835), confirming that local temporal supervision is necessary to avoid anatomical mismatch.

4.5 Discussion on Generalization

A limitation of this study is that quantitative evaluation is mainly conducted on Kvasir-Capsule [5]. Although this dataset covers diverse gastrointestinal appearances, validation on other endoscopic modalities, such as bronchoscopy and laparoscopy, is still needed. Since quaternion spectral modeling and temporal attention address general challenges in medical video restoration, we expect STQFormer to be extendable to broader clinical settings.

5 Conclusion

We proposed STQFormer, a novel spatio-temporal transformer framework that leverages quaternion algebra to preserve diagnostic color fidelity in endoscopic videos. By unifying spatial spectral modeling with efficient temporal attention, our method outperforms state-of-the-art baselines on the standard Kvasir-Capsule benchmark. Future work will extend this framework toward **real-time clinical deployment**, focusing on hardware acceleration and validation across diverse endoscopic modalities.

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